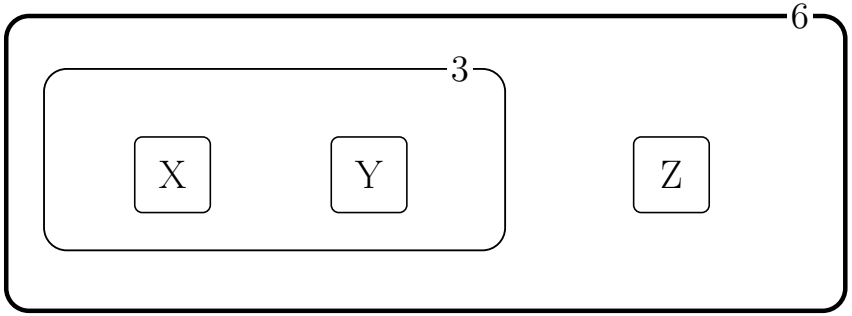
Groups can sit inside one another or be separate. They never cross. One group contains all the cards. Each card is also a one-card group. A group may contain three or more smaller groups. Every larger group has a positive label, and labels increase as you go outward. A group's size and number of cards do not determine its label.

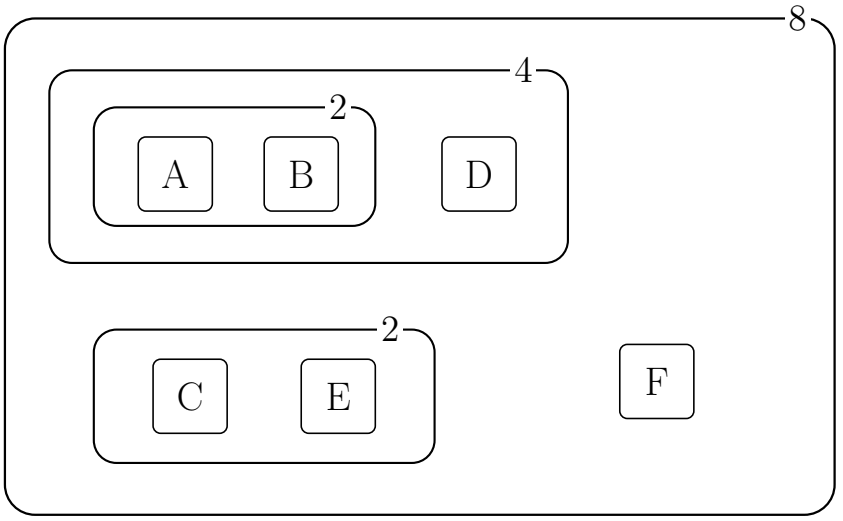
The distance between two different cards is the label of their smallest common group. A card's distance from itself is 0. We write A–B for the distance between A and B.

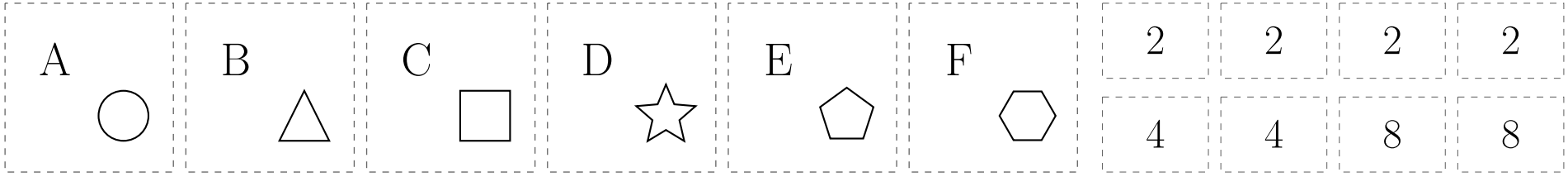


X and Z ⟶ the thick group is their smallest common group ⟶ X–Z = 6.

X–Y = 3. X–X = 0.

Problem 1: Build this world with cards A through F and loops. Choose six pairs and record their distances. Rearrange the cards and loops so every distance stays the same.





Problem 2: Use A, B, C and labels 2, 4, 8. Make each set of distances with loops, or cross it out if it cannot be made. Choose one crossed-out set and explain why no arrangement works.

A–B = 2 A–C = 2

B–C = 2

A–B = 2 A–C = 4

B–C = 4

A–B = 2 A–C = 2

B–C = 4

A–B = 4 A–C = 4

B–C = 4

A–B = 2 A–C = 4

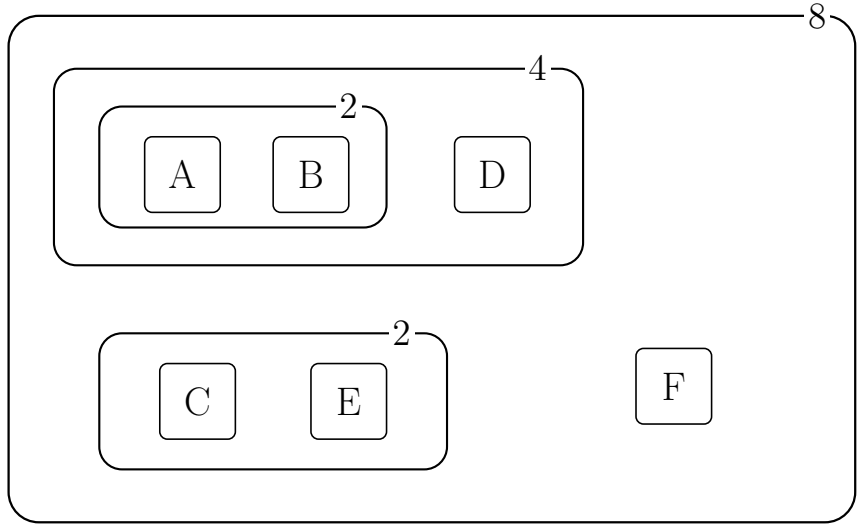
B–C = 8

A–B = 2 A–C = 8

B–C = 8

Problem 3: A ball of radius r around a card contains that card and every card at distance at most r. In the example on page 1, radius 1 around X gives just X; radius 4 around X gives X and Y.

In this world, find the radius-3 balls around A, C, D and F. Then find all different balls that contain B, using radii 0, 1, 2, 3, 4, 7 and 8. Draw each different collection once and write the radii that give it.



Problem 4: Use cards A through E to make a world that fits all ten clue cards. Use labels 2, 4, 8. Your partner checks each clue against the loops. Draw your world. Could loops around different sets of cards fit the same ten clues?

A–B = 2	A–C = 4	B–C = 4	D–E = 2	A–D = 8
A–E = 8	B–D = 8	B–E = 8	C–D = 8	C–E = 8